

Perovskite Photovoltaics in August 2026: Non-radiative recombination control and buried-interface treatment

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Abstract

Across a monthly literature corpus of 531 publications, 144 were examined closely to evaluate defect passivation, contact chemistry, phase stability, and operational durability in perovskite solar cells (PSCs). Independent certification establishes the performance frontier for monolithic tandem architectures at a power conversion efficiency (PCE) of 33.66%, driven by complementary bandgap absorption that suppresses thermalization losses alongside optimized current matching. Inverted single-junction configurations established a certified PCE ceiling of 27.4%, achieved primarily through tailored self-assembled monolayers (SAMs) and molecular additives that passivate undercoordinated lead defects and suppress interfacial molecular clustering. Scaling these passivated contacts to expanded dimensions incurs an area penalty governed by transparent conducting oxide sheet resistance, interconnect scribe dead area, and spatial film non-uniformity, yielding an accredited PCE of 17.35% at the largest reported aperture area of 61.2 cm². Operational longevity evidence remains sparse and disconnected from efficiency milestones: only 5 investigations quantified operational lifetime, with a maximum duration of 1400 h, while only 2 studies adhered to standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols. Degradation pathways identified across this cohort include interfacial redox reactions between acidic phosphonic acid anchoring groups and polar processing solvents, photo-induced halide oxidation and ion migration, and residual lattice microstrain that accelerates phase segregation under continuous operation. Reporting practices reveal a pronounced discrepancy between self-reported claims and certified performance, as 55.5% of abstracts stated an efficiency value, whereas only 9.7% stated independent certification and only 18 articles reported verified values. Although intermediate-phase modulation and seed-layer crystallization mitigate interfacial voids and non-radiative recombination at hole-selective contacts during initial deposition, published evidence fails to confirm that these strategies arrest field-driven halide redistribution or contact degradation during prolonged operation. Resolving these bottlenecks requires operando chemical characterization under bias combined with standardized ISOS testing on scalable apertures.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite;

tandem solar cells; operational stability

1. Introduction

Perovskite photovoltaic research has moved beyond maximizing unencapsulated power conversion efficiency (PCE) on sub-millimetre laboratory apertures. The strongest mechanistic consensus demonstrates that coordinating Lewis-base ligands and self-assembled monolayers (SAMs) passivate non-radiative recombination centres at buried contacts by coordinating undercoordinated lead sites. Yet the critical open problem has shifted to operational lifetime, device area, and rigorous verification. Arresting field-driven halide migration, phase segregation, and contact corrosion during prolonged continuous operation across scalable apertures remains unresolved, presenting barriers that transient efficiency improvements cannot circumvent.

To evaluate these challenges, 531 indexed works met the scope gate in August 2026 and 144 were read closely.

The resulting evidence base is structured across core physical domains: defect tolerance, passivation and ion migration, buried interfaces and contact chemistry, composition, phase control and crystallisation, wide-bandgap absorbers and tandem integration, as well as operational stability and degradation pathways.

Throughout each topic, this review distinguishes what the month’s literature verified from what it asserted. Theoretical passivation mechanisms and charge-transport claims are scrutinized directly against empirical device behavior, separating documented operational survivability from unverified assertions based on uncertified, small-area cells subjected to brief, non-standardized aging routines.

2. The Certified Frontier in August 2026

Independent certification establishes that spectral splitting across complementary bandgaps yields the highest verified efficiencies, driven by suppressed thermalization and optimized current matching in two-terminal (2T) perovskite tandems. Verified 2T power conversion efficiency (PCE) benchmarks reached a ceiling of 33.66%, followed by certified reports at 32.56%, 30.77%, 29.7%, and 27.84% [1,2,3,4,5]. Single-junction inverted (p-i-n) devices, constrained by non-radiative interfacial recombination and extraction barriers at selective contacts, established a narrower certified ceiling between 26.77% and 27.4%, led by verified values of 27.4%, 27.2%, and 26.77% [6,7,8]. Despite these verified milestones, rigorous certification remains rare across the literature: only 18 of 144 closely read papers reported an independently certified efficiency, allowing uncertified self-reported champion values to run ahead of verifiable baselines.

Translating these contact interfaces across larger dimensions exposes severe transport bottlenecks governed by transparent electrode sheet resistance, scribe line geometric dead area, and spatial film non-uniformity. At expanded dimensions, minimodule architectures maintained reported efficiencies of 27.7% on 25 cm² and 27.01% on 58.51 cm² [9,10]. However, larger apertures revealed significant performance losses, dropping to 23.3% on 57.3 cm², 23.14% on 29.7 cm², and 17.35% on 61.2 cm² [11,12,13]. A champion value recorded on a sub-0.1 cm² laboratory cell isolates microscopic interfacial transport under near-ideal current extraction, whereas an independently certified value measured across a module aperture gauges lateral conduction, spatial defect distributions, and coating reproducibility. These two metrics represent entirely distinct categories of physical evidence, meaning small-cell records cannot demonstrate module viability.

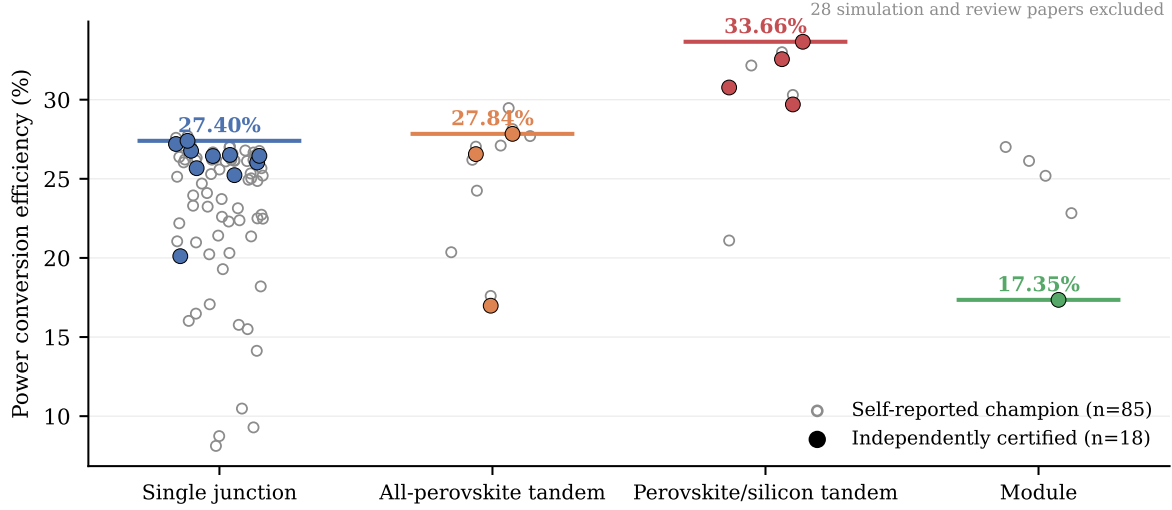


Figure 1. The certified frontier in August 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

3. Defect Tolerance, Passivation and Ion Migration

Defect formation and ion migration in metal halide perovskites are coupled through mobile halide vacancies that trigger operational degradation. Continuous illumination accelerates iodide oxidation to molecular iodine, generating iodine vacancies that expose undercoordinated lead (Pb^{2+}) sites [14], whereas nonadiabatic molecular dynamics shows that iodine vacancies drive hydrogen vacancy formation into stable VI · VH defect complexes whose migration is immobilized by rigid Pb-C coordinate bonding in VI · VH-C configurations [15]. Kinetic and compositional regulation confirms that mobile halide management is prerequisite to suppressing ion drift [16,17]. Exploiting the kinetic accessibility of bromide (Br^-) diffusion, an ordered anion-cation modification (OACM) strategy repairs anionic vacancies before cation passivation, suppressing phase segregation to achieve a 1.33 V open-circuit voltage (V_{oc}) and 19.21% stabilized efficiency in 1.83 eV wide-bandgap (WBG) top cells (champion 26.2% tandem efficiency) [16]. Selenocyanate pseudohalide alloying similarly retards ion drift and homogenizes formamidinium (FA^+) distributions throughout 1.66 eV absorbers [17]. Structural barriers provide complementary confinement: scaffold-directed polyfluorinated 7F-EA-HI forms an interwoven tilted one-dimensional/three-dimensional (1D/3D) heterointerface that blocks defect propagation and ion migration in inverted (p-i-n) devices reaching an uncertified 24.94% power conversion efficiency (PCE) [18], whereas self-driven halide exchange with bromide-rich nanocrystals passivates iodide vacancies via a surface-graded heterojunction, though restricted to an uncertified 22.19% champion PCE in an unreviewed preprint [19].

Beyond mobile halides, eliminating deep non-radiative traps demands coordinated passivation of undercoordinated metal cations and organic cation vacancies [14,20,21,22]. Bulk phenylthiourea coordinates Pb^{2+} to retard ion migration and suppress defect formation under illumination [14]. Ammonium alginate suppresses FA^+ deprotonation while passivating undercoordinated Pb^{2+} , raising fill factor (FF) from 80.2% to 85.1% in an uncertified 26.12% champion cell [20]. Dual-site coordination similarly targets opposing charges: sodium cyclamate coordinates Pb^{2+} via sulfonic acid groups and binds iodide vacancies through amino hydrogen bonds [21], while multidentate sulfamethoxa-

zole coordinates Pb^{2+} , organic-cation, and halide vacancies, lifting normal (n-i-p) efficiency from 18.01% to an uncertified 21.05% champion (20.31% average) [22]. Coupled hydrogen-bonding and electrostatic networks from the ionic liquid BSPH coordinate FA^+ , halides, and Pb^{2+} , suppressing trap density and ion drift to yield a 23.72% p-i-n PCE [23]. In hole-conductor-free carbon-electrode mesoscopic cells, DMSC passivates both Pb and I species to reach a 22.5% champion PCE [24], while multi-pyridyl spirobifluorenes coordinate defects during crystallization to yield 20.98% PCE versus 18.67% in controls [25]. In tin-based absorbers prone to Sn^{2+} oxidation, triphenylphosphine oxide coordinates Sn^{2+} as a hard Lewis base, yielding 14.13% PCE and a T80 of 720 h under unlabelled testing conditions [26].

A third frontier links defect formation directly to lattice mechanics and buried transport contacts [27,28,29]. Solid-solution lattice hardening by partial lanthanum substitution of Pb raises the Young modulus from 27.96 to 35.71 GPa, demonstrating that rigidifying the inorganic cage suppresses defect formation to yield a 26.54% champion PCE and a certified steady-state efficiency of 26.03% [27]. Dynamic strain homogenization via in situ supramolecular networks modulates Sn^{2+} coordination in mixed tin-lead absorbers, suppressing segregation to retain 84.7% initial efficiency after 450 h maximum power point tracking (MPPT) under one sun and 92.1% after 50 thermal cycles (24.1% champion PCE) [28]. At buried contacts, Ace-K bidentately chelates uncoordinated Sn^{4+} and oxygen vacancies on tin dioxide (SnO_2) while outcompeting dimethyl sulfoxide (DMSO) for lead iodide (PbI_2), eliminating voids to retain 92.4% efficiency after 1000 h MPPT on a 0.0729 cm^2 cell (26.17% champion PCE) [29]. Interfacial molecular bridging with 2-formylbenzenesulfonic acid sodium salt connects self-assembled monolayers (SAM) to perovskite, delivering an uncertified 26.21% champion PCE with an 86.15% FF [30]. At module scale, potassium dihydrogen phosphate disaggregates SnO_2 colloids and passivates buried contacts, yielding 23.16% PCE on 12 cm^2 modules with 8% scaling loss from a 25.34% champion cell [31].

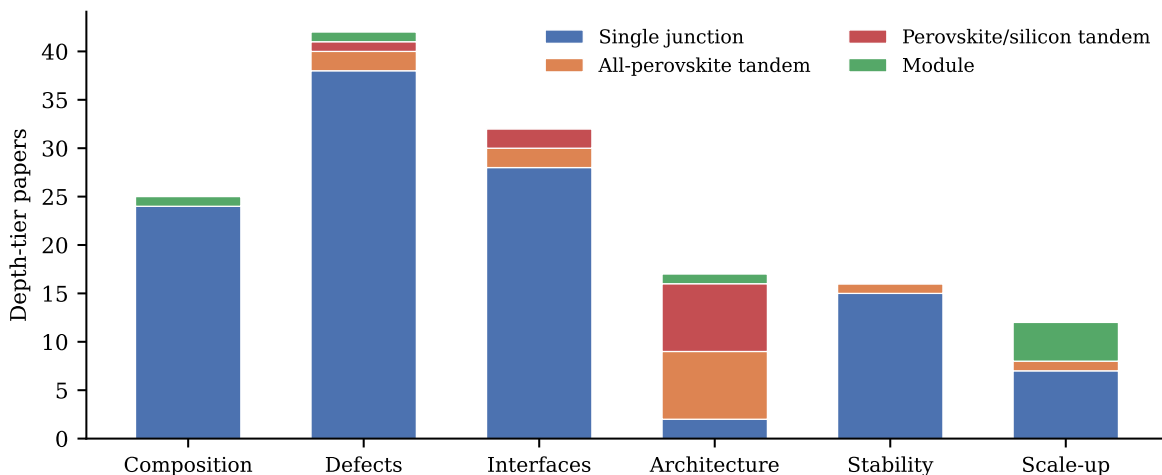


Figure 2. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

4. Buried Interfaces and Contact Chemistry

At buried hole-selective contacts in inverted positive-intrinsic-negative (p-i-n) and tandem perovskite solar cells (PSCs), disordered molecular packing and aggregation in self-assembled monolayers

(SAMs) cause severe pinhole shunts and energy misalignment. Suppressing self-assembled monolayer (SAM) clustering while keeping anchoring sites accessible resolves these losses: non-anchoring additives like TTMB secure open binding sites to reach a certified power conversion efficiency (PCE) of 33.66% (33.66% stabilised) in two-terminal (2T) perovskite/silicon tandems, hydrogen-bond networks via glycineamide hydrochloride in [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-4PACz) yield 22.38% champion PCE in pure-iodide wide-bandgap cells, and DMPA-based multilayers achieve a certified 27.2% PCE (27.59% champion) with 94.5% retention after 1600 hours under continuous 1 Sun illumination at 65 °C [1,7,32]. Interfacial dipole engineering further optimizes band offsets: depositing bismuth oxychloride (BiOCl) onto SAMs induces dipole flipping and chlorine-iodine exchange, yielding a certified 27.84% 2T tandem PCE (28.14% champion, 27.86% stabilised), although single-junction efficiencies of 27.25% (1.55 eV) and 20.24% (1.85 eV) remain uncertified on small 0.09 cm² aperture areas [5]. This aligns with co-assembling 4-ASA with Me-4PACz on nickel oxide (NiO_x) to yield a 20.36% four-terminal (4T) tandem PCE (17.82% scanned single junction), and dipole tuning via m-FBC raising PCE from 23.05% to an uncertified 25.03%, though stability was tested only under 500 hours of uncharacterized storage [33,34]. Asymmetric heteroatom conjugation also regulates contact energetics: S-AS on NiO_x achieves 25.70% steady-state output (26.49% champion) and LW-1 on fluorine-doped tin oxide (FTO) reaches 26.0% stabilised efficiency (26.38% champion) [35,36].

Beyond electrostatic alignment, buried contact molecules template crystallization, relieve lattice strain, and passivate undercoordinated lead (Pb²⁺) defects. Multi-arm donor-acceptor dipoles (T4IA) coordinate Pb²⁺ through C=N and methoxy groups to relieve residual strain (26.79% champion PCE), whereas non-covalent templating between precursor additives (TPANI) and structurally matched triphenylamine bisphosphonic acid SAMs on indium tin oxide (ITO) reduces intergranular groove depth and reinforces adhesion (26.58% champion) [37,38]. Interfacial Pb-S bonding provides a parallel passivation route: thienyl-based SAM additives template preferential (111) crystallization and segregate at grain boundaries to secure a certified 32.56% (33.0% champion) 2T tandem PCE, mirroring solution-phase sulfidation with dibutyl sulfide that induces an interfacial p-to-n transition to yield an uncertified 26.28% PCE [2,39].

Contact degradation also stems from chemical reactivity across metal oxide substrates. On NiO_x hole transport layers (HTLs), removing deleterious surface hydroxyls with ammonium thiocyanate (NH₄SCN) lowers transfer barriers and lattice strain (21.41% champion) [40]. Decoupling SAM anchoring from charge extraction through composite NiO_x stacks (sputtered base plus solution-deposited hydroxyl overlayer) permits scalable processing, achieving 24.86% PCE across 12.4 cm², consistent with enamine SAM anchor engineering that achieves 23.14% PCE with 83.11% fill factor (FF) on a 29.7 cm² mini-module (>25.5% on small cells) [12,41]. At tin oxide (SnO₂) electron transport layers (ETLs) in regular negative-intrinsic-positive (n-i-p) devices, native surface hydroxyls and oxygen vacancies provoke interfacial void formation and microcracking; these are suppressed either by converting hydroxyls into covalently anchored acetate via ethanol vapor reconstruction (26.19% champion) or by subliming pyruvic acid at 145 °C to purge defect sites, reaching 26.17% PCE on 1 cm² with an 80% operational lifetime (T₈₀) of 900 hours [42,43]. However, interfacial engineering reveals a trade-off between extraction and mechanical integrity: although fullerene SAMs (C₆₀-PA) on SnO₂ raise initial tracked efficiencies from 18.6% to a 20.0% stabilised value, performance degrades under illumination at elevated temperatures because fullerenes weaken adhesion at the oxide interface [44].

5. Composition, Phase Control and Crystallisation

Inhomogeneous nucleation triggers point defects, phase segregation, and structural disorder in metal halide perovskite absorbers, requiring precise crystallization modulation [45]. Controlling intermediate solvent-solute coordination dictates lattice orientation and residual stress: destabilizing solvent-PbI₂ adducts through cooperative interactions between sodium hydroxyquinoline sulfonate and sodium aminonaphthalene sulfonate drives out-of-plane [100] growth while converting tensile strain into compressive strain, yielding an uncertified champion power conversion efficiency (PCE) of 26.21% [46]. Similarly, dual-mode coordination with 4-chlorophenethylammonium (4Cl-PA) generates a macroporous PbI₂ template that facilitates organic cation infiltration, extending carrier lifetime to 161 ns by time-resolved photoluminescence (TRPL) in uncertified 22.73% efficient devices [47]. In tin perovskites, intermediate-mediated stepwise crystallization during a pre-annealing static stage with fluorinated phenethylammonium alters the [SnI₆]⁴⁻ electronic landscape to suppress Sn²⁺ oxidation, reaching a 9.29% PCE [48]. However, chemical coordination efficacy depends on the processing solvent: while SnF₂ suppresses Sn²⁺ oxidation in dimethylformamide:dimethyl sulfoxide (DMF:DMSO), it offers limited antioxidant protection in DMSO-free DMF:2-methylaminoethanol (DMF:2MAP), where formamidinium tin iodide (FASnI₃) films remain inherently Sn²⁺-rich [49].

Phase stability is equally bounded by thermodynamic competition and local defect equilibria [50,51,52]. In cesium-based systems, excess PbI₂ suppresses Ruddlesden-Popper antiphase boundaries (RP-APBs) but triggers edge-sharing [PbI₆]⁴⁻ motifs that nucleate the non-photoactive delta phase; dimethylammonium (DMA⁺) phase engineering circumvents this breakdown by forming beta-(DMA,Cs)PbI₃ and intrinsically eliminating RP-APBs to achieve an uncertified 20.23% PCE [50]. Similarly, incorporating 0.8 mol% fluorinated methylammonium iodide (FMAI) stabilizes the perovskite lattice against thermal degradation in a 25.67% negative-intrinsic-positive (n-i-p) device, though the reported 2000 h operational stability lacks detailed testing protocols [51]. In low-dimensional heterostructures, organic cation evaporation during annealing drives a dimensional shift from low- to high-n quantum wells, where precursor octylammonium bromide (OABr) concentration predetermines the initial n = 1 phase distribution in 19.29% wide-bandgap cells [52]. Replacing insulating organic spacers with semiconducting diketopyrrolopyrrole cations enables bidirectional charge transfer with inorganic [PbI₄]²⁻ slabs and yields electron mobilities up to $1.3 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, although functional solar cell metrics were omitted [53].

In lead-free multinary lattices, accessing target crystal phases requires overcoming severe kinetic and thermodynamic barriers [54,55,56]. Sequentially evaporating methylbismuth iodide (MBI) bilayers at substrate temperatures as low as 40 °C requires post-deposition annealing above 200 °C to yield a monoclinic C2/c phase with (111) texture, yet champion PCE reaches only 0.53% [54]. In double perovskites, solid-state mechanosynthesis reveals strict ionic size selectivity: liquid assistance drives Rb⁺ into the A-site and Na⁺ into the B-site of Cs₂AgInCl₆ [55]. Conversely, solvent-free hydro-solution precipitation directly produces cubic Fm-3m Cs₂AgBiBr₆ without high-temperature annealing [56], indicating that solution-phase ionic transport bypasses the kinetic barriers inherent to vapor-phase and solid-state transformations.

6. Wide-Bandgap Absorbers and Tandem Integration

Non-conformal coverage and dewetting of hole-selective self-assembled monolayers (SAMs) on textured silicon substrates generate interfacial voids and unreacted lead iodide (PbI₂), driving non-radiative recombination in two-terminal (2T) perovskite/silicon tandems [4,57,58]. Resolving this

buried morphological mismatch requires tailored nucleation and crystallization: a CsCl seed layer facilitates organic precursor incorporation across MeO-2PACz to suppress interfacial PbI_2 , yielding a certified 29.7% power conversion efficiency (PCE) [4], while solvent-free sequential evaporation capitalizes on accelerated lead halide conversion on textured silicon to deliver a 27.0% stabilized PCE [59]. Post-deposition methylammonium thiocyanate (MASCN) healing similarly repairs valley-confinement defects by driving secondary crystal growth into full-thickness columnar grains, achieving a certified 30.77% PCE on a 1.164 cm^2 aperture and retaining initial performance after 3400 hours of continuous ambient maximum power point tracking (MPPT) [3]. In the wide-bandgap (WBG) top absorber, phase segregation and halide migration govern bulk degradation. Elevating the ion migration energy barrier improves local structural stability, supporting an uncertified champion PCE of 33.02% [60], while two-dimensional (2D) (APD) PbI_4 seed templates impart regular compressive strain into the three-dimensional (3D) lattice to suppress phase separation for a 29.47% PCE [61]. Contact durability under subsequent processing presents a further constraint: sacrificial HTFNA coordination shields SAMs against atomic layer deposition (ALD) damage, yet this certified 26.56% efficiency is restricted to a 0.062 cm^2 cell [62], highlighting the shift toward resilient inorganic contacts like nickel oxide with N719 dye that preserve 96.4% PCE after one year of outdoor MPPT [63]. Translating these interfaces to large-area modules escalates multi-scale defect densities [64], requiring kinetic crystallization control: polyfluoroarene additives elevate effective nucleation barriers to yield a 27.01% module PCE across 58.51 cm^2 with 55 mV non-radiative voltage loss [10], matched by static antisolvent engineering achieving 27.7% on 25 cm^2 [9].

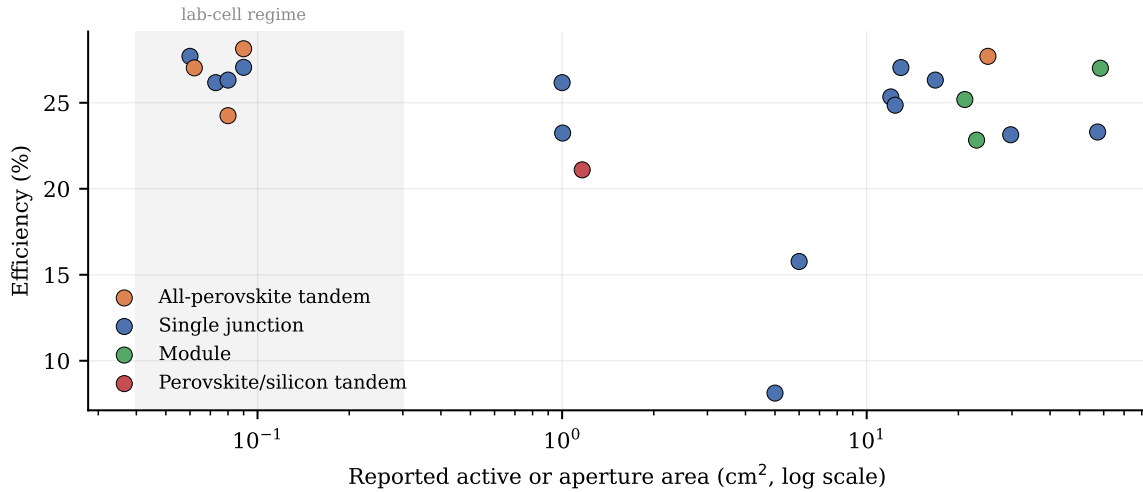


Figure 3. Efficiency against reported active or aperture area. The shaded band marks the sub- 0.3 cm^2 laboratory-cell regime in which most champion values are measured.

7. Operational Stability and Degradation Pathways

Interfacial chemical and photochemical degradation at selective contacts constitutes the primary failure mechanism addressed this month. Inverted perovskite solar cells (PSCs) suffer from coupled redox degradation: acidity in carbazole-based phosphonic acid self-assembled monolayers (SAMs) activates dimethyl sulfoxide (DMSO)-mediated iodide redox reactions during ink deposition [6], while photo-oxidation of formamidinium iodide (FAI) yields iodine radicals that catalyze phenyl-C61-butyric acid methyl ester (PCBM) dimerization alongside FAI deprotonation [65]. Cross-linking self-assembled

monolayer (SAM) headers in situ [66], co-assembling binary monolayers [67], or balancing dipole configurations [68,69] suppresses interfacial defect generation. Yet performance claims frequently exhibit severe certification discrepancies or rely on tiny device areas, such as a 27.06% uncertified champion dropping to 20.11% certified on 0.09 cm^2 [69] and a certified 27.4% power conversion efficiency (PCE) tested on 0.06 cm^2 [6].

In the bulk absorber, unrelaxed microstrain and defect formation accelerate phase degradation. Relieving lattice strain through 2 mol% sodium (Na^+) interstitials [70], two-dimensional (2D) intermediate phase templating [71], or vertical lattice misalignment suppression [72] increases defect formation energies for lead and iodide vacancies and prevents nonphotoactive phase transitions [70,71,72]. At rear contacts, atomic interdiffusion is curtailed by inserting a 5 nm chromium (Cr) barrier beneath non-gold electrodes [73] or using coordination modifiers to prevent direct perovskite contact with metal oxides [74].

Standardized operational metrics remain conspicuously scarce. Across 144 examined papers, only 5 reported a time to 80% of initial efficiency (T80) and merely 2 adhered to International Summit on Organic Photovoltaic Stability (ISOS) protocols, specifically ISOS-L-1 [67] and ISOS-L-2 [65]. Consequently, isolated metrics such as an 800 h T80 [70] or an 864 h T80 on 0.08 cm^2 [74] reflect unstandardized evaluations rather than verified operational longevity.

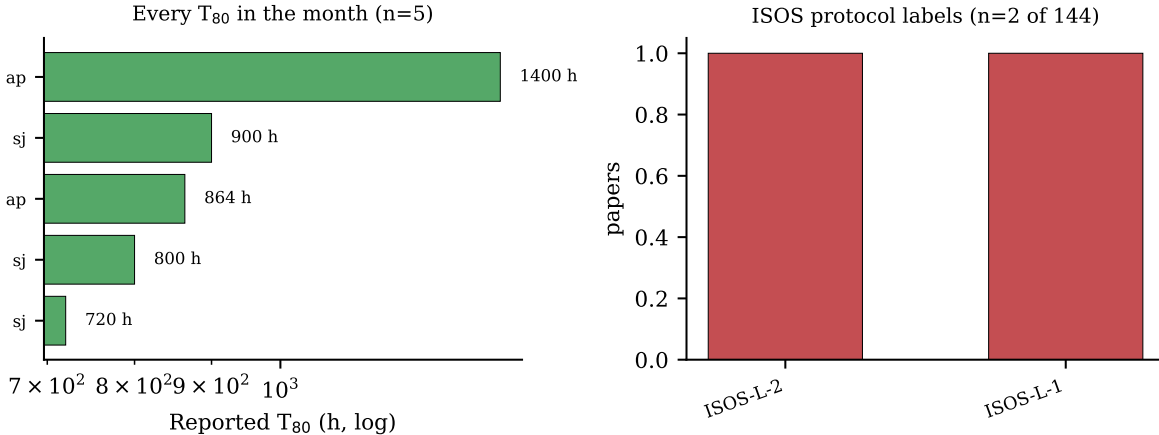


Figure 4. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

8. Research Gaps and Outlook

Controlling wide-bandgap (WBG) perovskite crystallization across textured silicon substrates represents the most supported mechanism this month, advancing certified two-terminal (2T) power conversion efficiency (PCE) to 33.66% [1] and 30.77% [3]. However, the assembled evidence leaves critical physical questions unestablished across four distinct areas.

First, although seed layers and secondary grain growth suppress phase segregation during deposition [3,4], current data do not demonstrate that columnar grains prevent electric-field-driven halide redistribution under sustained bias. Operando depth-profiling time-of-flight secondary ion mass spectrometry (ToF-SIMS) under operational voltage bias is required to determine whether seeded interfaces permanently impede ion migration or merely delay initial demixing. Second, while non-

anchoring additives reduce aggregation in self-assembled monolayers (SAMs) [1], the literature fails to verify whether these modified contacts survive wet chemical processing: acidic phosphonic acid SAMs trigger dimethyl sulfoxide (DMSO)-mediated iodide redox reactions during ink coating [6], leaving uncertified champion efficiencies vulnerable to severe certification discrepancies [69]. In situ grazing-incidence wide-angle X-ray scattering (GIWAXS) coupled with interface-sensitive photoelectron spectroscopy during solution wetting must establish whether molecular additives protect buried contacts from chemical degradation. Third, device scalability remains disconnected from small-area interfacial gains: scale-up was addressed by 27 of 531 works, with the largest reported aperture reaching 61.2 cm² [13], yet reported data do not decouple the physical origins of area-dependent efficiency penalties [71]. Calibrated absolute electroluminescence imaging resolving spatial ideality factors across square-decimeter modules is necessary to separate interconnect sheet resistance from spatial non-radiative recombination. Fourth, operational longevity remains unverified across the field. Only 5 papers reported a time to 80% of initial performance (T80) lifetime, with the longest reaching 1400 h [75], and only 2 papers named an International Summit on Organic Photovoltaic Stability (ISOS) protocol [65,67]. Standardized ISOS testing combining elevated thermal stress with continuous maximum power point tracking (MPPT) under full illumination is needed to isolate contact delamination from intrinsic lattice decay.

The certified reporting share was 8.3 in 2026-07 and 9.7 in 2026-08, yet only 18 of 144 closely read papers reported any certified value. Requiring standardized ISOS protocols alongside certified aperture-area efficiencies would make next month’s literature directly comparable across laboratories.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

AI Usage Declaration

This manuscript was prepared as part of our Research Project for making a reliable Autonomous Researcher Agent. The agent executed the monthly review pipeline without human intervention: it harvested the indexed literature for the month through the OpenAlex, Semantic Scholar, Crossref and arXiv interfaces, applied a deterministic topical filter, ranked and selected the papers read closely, extracted every quantitative claim into a structured record binding each number to a verbatim quotation from that paper’s own abstract, computed the statistics and figures from those records, and drafted this text from the resulting evidence briefs. Literature harvesting, selection, statistics, figures and gate checking are deterministic scripts rather than model output. Structured

extraction used qwen3.8-flash through the opencode interface; the prose was written by gemini-3.8-flash-high through the agy interface; independent review used Opus 5 at high reasoning effort. No figure is a model-generated image: every panel is plotted directly from the extracted records. Every quantitative statement in the main text resolves to the machine-generated statistics file shipped as Supplementary Information, and every citation resolves to a record carrying its source quotation. The author reviewed and edited the content, verified the selection, and takes full responsibility for the content of the published article.

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